Brief

Data has become an important asset for companies and the way they operate. Moreover, data transmission and networks have enabled enterprises to improve their businesses, foster collaboration, increase their efficiency, enhance customer engagement and so on.

Based on the knowledge gained in the first five weeks in this subject, as well as your own research, you need to come up with a report describing the network of the chosen company, and its main characteristics, as required in the brief.

Select a company that you have recently interacted with, which is basing some of its operations on network and data. You need to be familiar with some of their activities in the digital space, based on your own experience and/or your own research. The company can be a medical centre, the public transport company you are using every week, your insurance company, etc.

Alternatively, you may use the list below to choose a suitable option: https://brandirectory.com/rankings/australia/table

A full network diagram with all the adequate network devices has to be created to support your discussion.

Learning Outcomes:

The Subject Learning Outcomes demonstrated by the successful completion of the task below include:

- Describe and analyse the architecture and distribution of data.
- Compare and review communication protocol stacks and generate recommendations.

This assessment gives you the opportunity to apply the knowledge and theories you've learnt in this subject. In doing so, you are required to analyse the chosen company's data distribution. Provide reviews and recommendations of the data flows and network communication protocols.

Please refer to the Task Instructions (below) for further details on how to complete this task.

Instructions/Requirements

1. Once you have chosen your company, create a Network Diagram representing its network, and describe it briefly.

    a. What are the main devices for the network? Consider networking devices, internet access, connections, data storage, data access devices, etc.
    b. The diagram has to be your authentic work, and you need to create it using tools such as **Packet Tracer**. Please save your draft in the application you have used. Diagrams from the web won't be accepted.
    c. The diagram has to reflect the chosen company. For instance, if the company has different branches, departments or warehouses, you should include those characteristics there. Or, if the company is using specific devices (e.g. self-checkout machines, IoT devices, sensors, scanning devices, etc.), you should include them.
    d. You may make reasonable assumptions for the inclusion of certain devices in the network.
    e. All the diagrams/pictures used must be referenced according to APA 7 referencing style.
    f. This is not a business diagram, logical diagram, or flowchart. Instead, this is a Network Diagram.

2. Choose one specific interaction between you and the company, or between you and an employee, or between the employee of the company and its system, where data is transmitted over the network. Think about the flow of data (data trail).
    a. What are the devices involved in data creation, transmission and storage?
    b. What type of media communications are utilised throughout the end-to-end data transmission? Please list a couple.
    c. What are some of the protocols used in various steps? Why are they used? Provide at least 3 relevant protocols. Please support your discussion with external references.

3. Provide three (3) recommendations to the company, in relation to data distribution and data safety, that you, as a Network Administrator, think (from your research) that the company should act. The recommendations should be around the network technology used, protocols, network architecture, etc. The recommendations should be short, sharp and actionable. Please support your discussion with external references.

Suggested Submission Guidelines

1. Cover page
2. AI declaration form (on a separate page)
3. Table of Contents (on a separate page; The ToC should be created through MS Word)
4. Introduction – provide an overview of the company.
    a. Not more than three (3) sentences about the chosen company.
    b. In brief – what is your work about?
    c. What the reader can expect to find in the body of the report
5. Discussion – *The three sections are provided in the "Task Instructions" above.*
    a. Use proper headings to identify **each** of the sections.
6. Conclusion – a summary of your findings. Do not introduce any new material in your conclusion.
7. References list – this starts on a new page. Please refer to APA 7th edition. **All** the references in the list must **match in-text references**.
8. Referencing
    a. Please use between 5-7 references including textbooks or academic journals.
    b. It is essential that you use the appropriate APA 7 style for citing and referencing research. Please see more information on referencing in the https://library.torrens.edu.au/academicskills/apa/tool.